ANALYTICS PLAN

CANDIDATE	ROLE	COMPANY
Abhinav Chaudhary	Growth Product Manager	Bhanzu

Covers the events, metric formulas, data quality checks and privacy rules for the Math Mission experiment. Each event is marked by where it lives today:

- **P — implemented in the prototype** (fires in the browser, visible in the Event log, never leaves the device)
- **B — needs a backend or external integration** (booking system, WhatsApp provider, class platform, CRM, payments)
- **R — required before a production experiment** (applies to every B event and to P events, which must move to a real pipeline)

1. IDENTIFIERS

ID	WHAT IT IS	USED FOR
`booking_id`	Pseudonymous ID from the booking system (prototype: `bk_sample_xxxxxx`)	Joins every event for a booking; the unit of analysis
`household_key`	Salted SHA-256 of the normalized parent phone number	Randomization and deduplication only; never stored with names
`event_id`	UUID generated per event	Deduplication
`session_id`	Random ID per device (prototype: stored in the browser)	Sessionizing views
`experiment_arm`	`control` or `treatment`, from assignment	Every event carries it (prototype always `treatment`)

The child's name, parent's name, phone and email are **never** event properties. The prototype enforces this: properties whose names suggest personal data are stripped before an event is recorded (`sanitizeProps`, unit-tested), and the browser test checks the event log never contains the child's name.

2. EVENT ENVELOPE

Every event has:

FIELD	RULE
`event_id`	UUID; server drops repeats
`name`	From the list below
`ts`	UTC, ISO-8601 with milliseconds
`tz`	Device IANA time zone (e.g. `Asia/Kolkata`); used to compute local-time features such as "booked in the evening"
`schema_version`	Integer; bump on breaking change (prototype: 2)
`props.booking_id`, `props.experiment_arm`, `props.session_id`, `props.booking_gap_days`	Added to every event

Time rules. Store everything in UTC. Compute day gaps in the **family's** time zone as calendar days (booked Monday 11 pm, demo Tuesday 9 am = 1 day), because that matches how families experience the wait. Server-side events (`booking_confirmed`, `demo_attended`) are the source of truth for the timestamps that define windows; client timestamps are used only for ordering within a session.

3. EVENTS

EVENT	STATUS	TRIGGER	REQUIRED PROPERTIES	DEDUP RULE	SOURCE OF TRUTH	METRIC USE	PRIVACY NOTES
`booking_confirmed`	B, R	Booking saved in the booking system	booking_id, household_key, child_age_band, booked_at, demo_at, gap_days, channel, campaign, market, slot_offered_first	One per booking_id	Booking DB	Eligibility, strata, denominator	Age band, not date of birth
`experiment_assigned`	B, R	Assignment at booking	booking_id, household_key, arm, stratum, salt_version	One per booking_id	Assignment service	Arm, sample ratio checks	—
`reminder_sent`	B, R	Provider accepts a template message	booking_id, template, message_no, arm	One per booking × template	WhatsApp provider	Exposure checks	Provider message ID only

EVENT	STATUS	TRIGGER	REQUIRED PROPERTIES	DEDUP RULE	SOURCE OF TRUTH	METRIC USE	PRIVACY NOTES
`message_delivered / message_read`	B, R	Provider webhook	booking_id, template, message_no	One per booking × template × status	WhatsApp provider	Activation denominator (Message 1 delivered); delivery parity by arm	—
`experience_viewed`	**P**, R	Hub or details page opened from the link	entry, hours_to_demo	First per booking per 30-minute session counts	Client → pipeline	Open rate (both arms)	—
`mission_viewed`	**P**, R	Mission start screen (age choice) shown	from	First per booking per session	Client	Funnel: viewed → started	—
`mission_started`	**P**, R	Age band chosen; first puzzle shown	age_band, attempt_number	**First per booking** counts for activation	Client	**24-hour activation** numerator	Age band only
`mission_resumed`	**P**	Continue after leaving or refreshing	age_band, question_index	—	Client	Content length diagnostics	—
`mission_answer_submitted`	**P** (prototype only, see §6)	Each answer	age_band, question_id, question_index, attempt, outcome	event_id	Client	Content difficulty	Child behavioural data; **not for production without legal sign-off**
`mission_completed`	**P**, R	Last puzzle resolved	age_band, first_try, solved, total	First per booking counts	Client	Completion rate; teacher/counsellor summary	Skills and outcomes only
`mission_retried`	**P**	Play again / other level	mode	—	Client	Engagement depth	—
`demo_details_viewed`	**P**, R	Prep page opened	from	First per booking per session	Client	Parent engagement	—
`joining_instructions_opened`	**P**, R	Join page opened	from	First per booking per session	Client	Readiness	—
`calendar_add_clicked`	**P**, R	Google Calendar link, copy details, or .ics	method	One per booking × method	Client	Readiness. **Does not mean an event was saved**	—
`readiness_item_checked`	**P**	Checklist item ticked	item	One per booking × item	Client	Readiness	—
`reschedule_started`	**P**, R	Change-time page opened	from	First per booking per session	Client	Conflict detection	—

EVENT	STATUS	TRIGGER	REQUIRED PROPERTIES	DEDUP RULE	SOURCE OF TRUTH	METRIC USE	PRIVACY NOTES
`reschedule_completed`	**P** (simulated), B, R	New slot confirmed	old_days_to_demo, new_days_to_demo, reschedule_count	One per booking × new slot	Scheduling system (prototype: simulated)	Reschedule rate; window handling	—
`demo_attended`	B, R	Class platform records the child joining	booking_id, join_offset_min, minutes_attended	One per booking (final scheduled demo)	Class platform	**Primary metric**	—
`demo_no_show`	B, R	Derived: scheduled demo ended with no join	booking_id, demo_at	One per booking × scheduled demo	Derived from class platform + schedule	Primary metric complement; triggers Message 5	—
`booking_cancelled`	B, R	Family or ops cancels	booking_id, reason_code	One per booking	Booking DB	Guardrail	Reason code, not free text
`counsellor_call_completed`	B, R	CRM logs a connected call	booking_id, connected, duration_bucket, summary_viewed	One per booking × call	CRM	Guardrail (workload); downstream	Call notes stay in the CRM
`enrollment_completed`	B, R	Payment succeeds	booking_id, plan, days_from_demo	One per booking	Payments	Guardrail / downstream	No payment details in analytics
`experience_reset`	**P** (prototype only)	Reset clicked	—	—	Client	None (reviewer tool)	—

4. METRIC FORMULAS

METRIC	FORMULA	NOTES
Attendance rate (primary)	bookings with `demo_attended` on the final scheduled demo within 14 days of the original slot ÷ eligible assigned bookings	Uses Bhanzu's existing "attended" definition; ITT
Meaningful attendance	as above with `minutes_attended ≥ 10`	Secondary
24-hour activation	bookings with first `mission_started` ≤ 24h after Message 1 `message_delivered` ÷ treatment bookings with Message 1 delivered and demo ≥ 24h after delivery	Leading indicator
Pre-demo activation (short notice)	bookings with `mission_started` before demo_at ÷ treatment bookings with < 24h from delivery to demo	Reported separately
Completion	bookings with `mission_completed` ÷ bookings with `mission_started`	Treatment
Open rate	bookings with `experience_viewed` ÷ eligible bookings	Both arms

METRIC	FORMULA	NOTES
Reschedule rate	bookings with ≥1 `reschedule_completed` ÷ eligible bookings	Both arms
Cancellation rate	bookings with `booking_cancelled` ÷ eligible bookings	Guardrail
Opt-out rate	bookings whose household opted out during the test ÷ eligible bookings	Guardrail

5. DATA QUALITY CHECKS (DAILY DURING THE TEST)

- Sample ratio mismatch on arm counts (chi-square, $p < 0.001$ pauses).
- Gap-bucket and market balance by arm.
- Message 1 delivery rate by arm (should be equal).
- Duplicate `event_id` rate; events with missing `booking_id`; events with timestamps in the future or before the booking.
- Client vs server clock skew on `experience_viewed`.
- `demo_attended` + `demo_no_show` + `booking_cancelled` covers every closed booking exactly once.

6. PRIVACY, CONSENT, RETENTION, ACCESS

This section describes design choices; legal compliance has **not** been established and needs review by counsel.

- **Children's data under India's DPDP Act.** The Act requires verifiable parental consent before processing a child's (under 18) personal data and prohibits tracking or behavioural monitoring of children and targeted advertising directed at them. The exemptions in the DPDP Rules are narrow and, per legal commentary, do not cover analytics or profiling (KS&K summary). Consequences for this design:
  - Production experiment analytics should be **booking-level**: started, completed, attended. Those describe the booking's progress, which the parent requested by booking.
  - **Per-question events** (`mission_answer_submitted`) are behavioural data about a child. They exist in the prototype to show how content difficulty could be measured. In production they should either be dropped or aggregated on the device into anonymous per-question counts with no booking ID, **subject to legal sign-off**.
  - The teacher/counsellor summary (skills that went well or needed help) is used only to deliver the class the parent booked, and only after consent captured at booking.
  - International markets (e.g. COPPA in the US, GDPR in the EU/UK) need separate review.
- **Consent:** parental consent and WhatsApp opt-in at booking; STOP honoured per booking.
- **Data minimisation:** first name for display only; age band instead of age; no school, location, photo, voice or free text.

- **Retention (*proposed*):** raw events 90 days; aggregated metrics kept; mission summary deleted 14 days after the final scheduled demo unless the family enrolls.
- **Access control (*proposed*):** teachers see only the summary for their own class bookings; counsellors through role-based CRM access; analysts see pseudonymous data only; the household key salt is held by the assignment service.
- **Honesty in the UI:** the prototype never says a message was sent, a calendar event created, a booking changed, or an event reported to a server. Each simulated action is labelled where it happens.